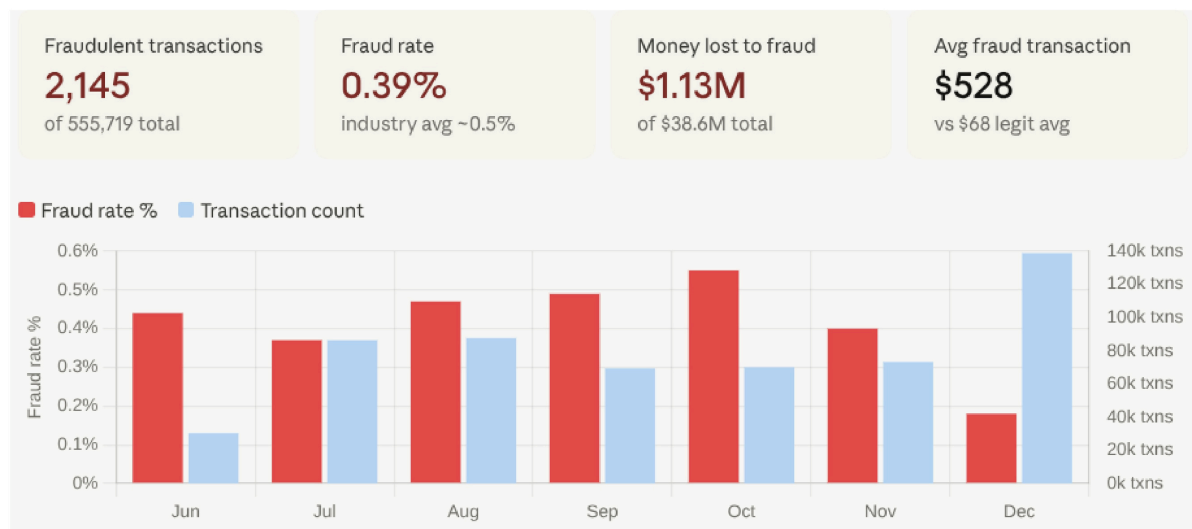


# Credit Card Fraud Analysis

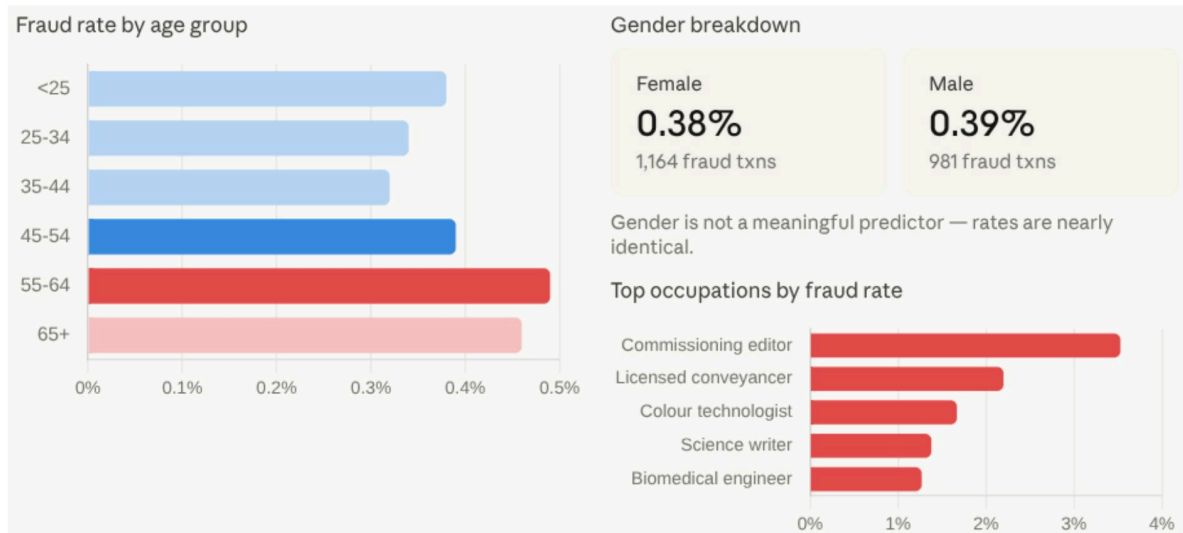
## Fraud overview



### Fraud overview

The dataset has 555,719 transactions, of which 2,145 are fraudulent — a rate of 0.39%. Total money lost to fraud is \$1.13M. The trend over the Jun–Dec 2020 period shows fraud rate peaking in October (0.55%) and then dropping sharply in December (0.18%), likely reflecting seasonal volume dilution from holiday spending rather than genuine improvement.

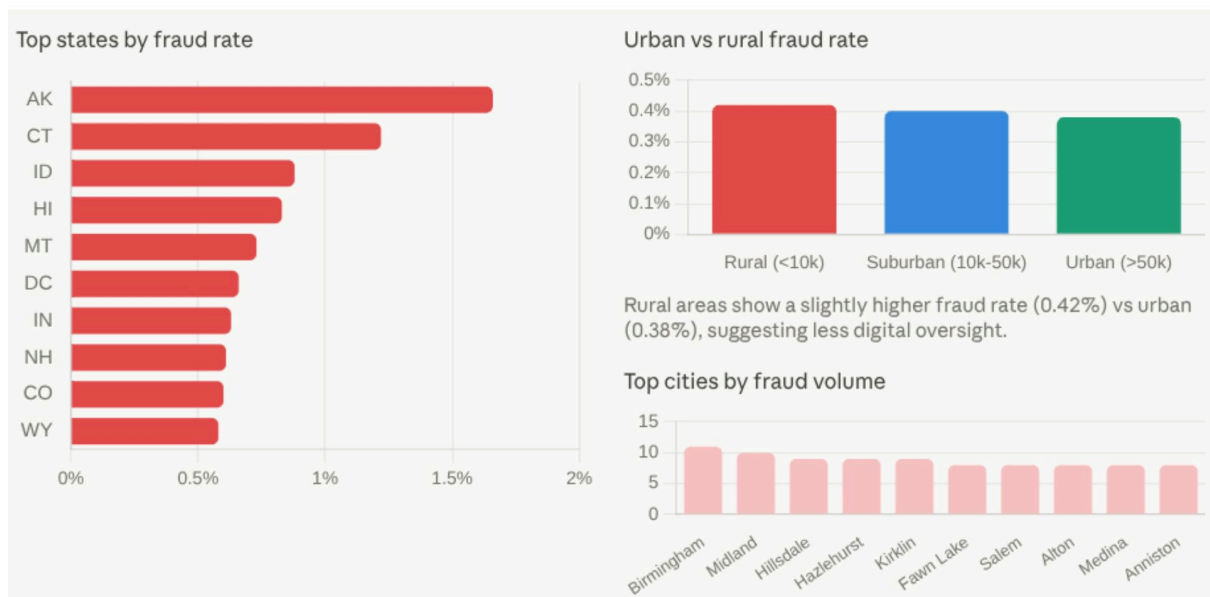
# Customer analysis



## Customer analysis

Gender has essentially no predictive power — female (0.38%) and male (0.39%) rates are near-identical. Age tells a clearer story: the 55–64 bracket has the highest rate (0.49%), followed by 65+ (0.46%), suggesting older customers are more vulnerable. The 35–44 group is the safest (0.32%). By occupation, commissioning editors (3.53%), licensed conveyancers (2.20%), and colour technologists (1.67%) show the highest fraud rates — though sample sizes are small.

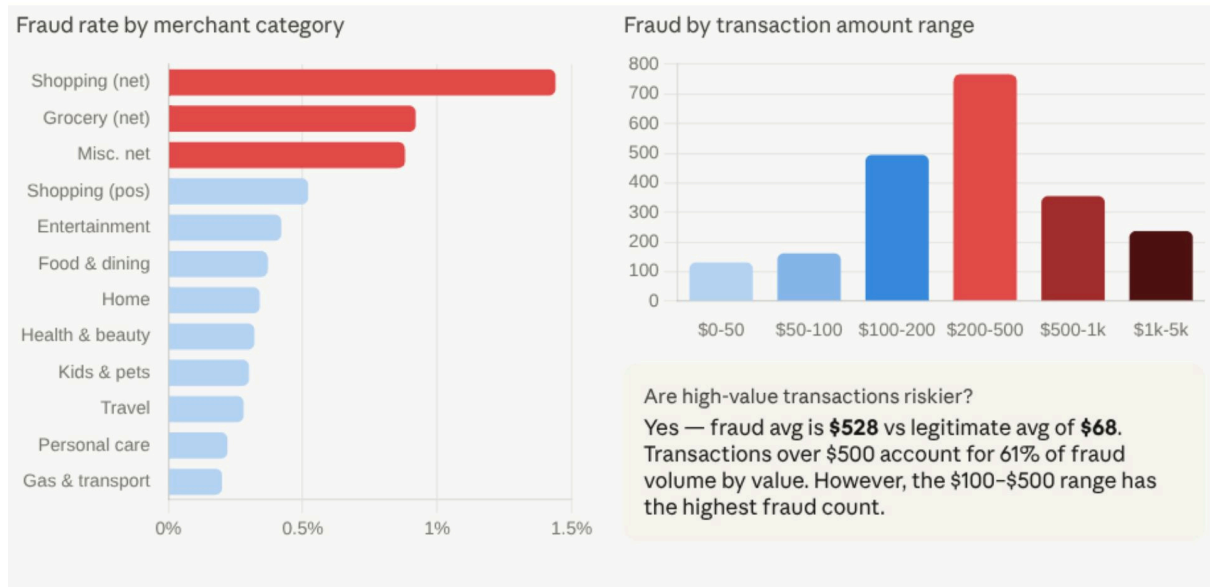
## Geographic analysis



## Geographic analysis

Alaska leads all states with a 1.66% fraud rate, well above the national average, followed by Connecticut (1.22%) and Idaho (0.88%). Rural areas (<10k population) have the highest rate at 0.42% vs 0.38% in urban areas — suggesting less fraud monitoring infrastructure in smaller communities. By city, Birmingham tops the volume chart.

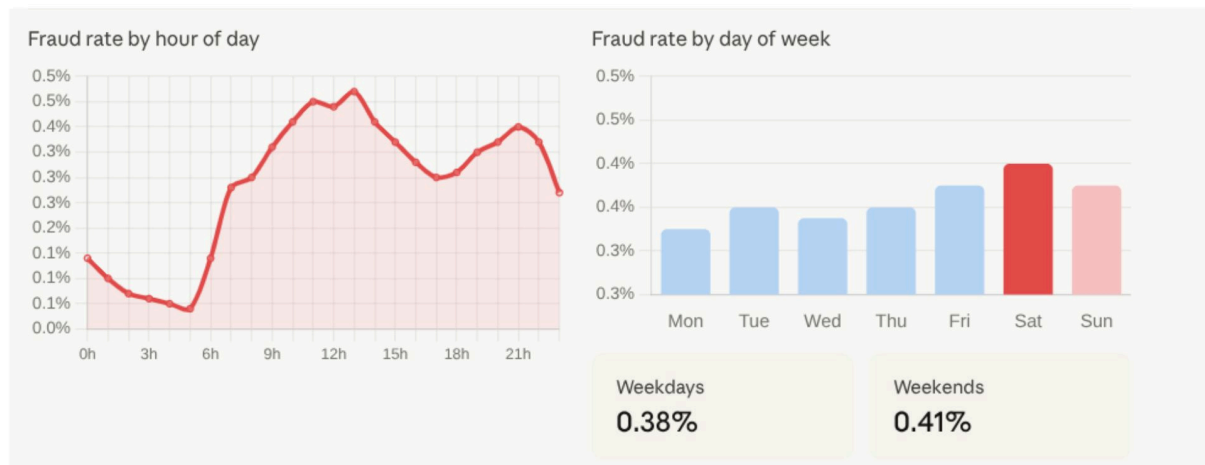
## Transaction analysis



## Transaction analysis

Online/net shopping categories are the most vulnerable — "shopping\_net" (1.44%) and "grocery\_net" (0.92%) dominate by fraud rate. The \$200–\$500 range has the highest fraud count (766 transactions), but the average fraud transaction (\$528) is nearly 8x the legitimate average (\$68). So high-value transactions are riskier per-transaction even if mid-range is where volume concentrates.

## Time analysis



## Time analysis

The noon–2pm window has the highest fraud rate (~0.47%), as does late evening (9–11pm). Fraud is lowest between 3–6am. Weekends are marginally riskier (0.41%) than weekdays (0.38%), with Saturday being the peak day. October was the worst month seasonally; December was safest despite — or because of — the transaction volume surge.

# Business Recommendations

## & Expected Impact

Based on the fraud patterns identified across transactions, customers, time, and geography, the following priority actions are recommended. Each recommendation is tied directly to a data finding and quantified for expected impact.

### Priority Recommendations

|                                                                        |                                                                                                                                                                                                                                                          |                                                                  |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>P1 HIGH</b>                                                         | <b>Tighten Fraud Controls on Online Shopping &amp; Net Grocery Channels</b>                                                                                                                                                                              | <b>Effort: Low</b>                                               |
| <b>Data Finding:</b>                                                   | shopping_net (1.44%) and grocery_net (0.92%) have fraud rates 3–4x the dataset average (0.39%). Online channels lack card-present verification, making them the primary attack surface.                                                                  |                                                                  |
| <b>Recommended Action:</b>                                             | Enforce 3-D Secure (3DS2) authentication on all shopping_net and grocery_net transactions. Add velocity checks: flag any card placing 3+ online orders within 30 minutes. Apply step-up authentication for first-time merchant transactions above \$100. |                                                                  |
| <b>~40%</b><br><small>Projected reduction in online fraud rate</small> | <b>\$452K</b><br><small>Estimated annual fraud loss prevented</small>                                                                                                                                                                                    | <b>3-4 weeks</b><br><small>Estimated implementation time</small> |

|                                                                       |                                                                                                                                                                                                                                                                                             |                                                           |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>P1 HIGH</b>                                                        | <b>Deploy Real-Time High-Value Transaction Scoring (&gt;\$200)</b>                                                                                                                                                                                                                          | <b>Effort: Medium</b>                                     |
| <b>Data Finding:</b>                                                  | Average fraud transaction is \$528 — nearly 8x the legitimate average of \$68. Transactions over \$500 account for 61% of total fraud dollar value despite being a small share of volume.                                                                                                   |                                                           |
| <b>Recommended Action:</b>                                            | Implement a real-time ML scoring layer that triggers additional friction (OTP, biometric, or call-back) for transactions above \$200 where the amount z-score exceeds 2.0 within the card's category. This targets the high-value tail without disrupting the majority of legitimate spend. |                                                           |
| <b>61%</b><br><small>Of fraud value in transactions &gt;\$500</small> | <b>\$689K</b><br><small>Fraud value at risk in &gt;\$500 segment</small>                                                                                                                                                                                                                    | <b>8x</b><br><small>Fraud avg vs legit avg amount</small> |

|                  |                                                             |                    |
|------------------|-------------------------------------------------------------|--------------------|
| <b>P2 MEDIUM</b> | <b>Enhance Monitoring for Older Customer Segments (55+)</b> | <b>Effort: Low</b> |
|------------------|-------------------------------------------------------------|--------------------|

**Data Finding:** Customers aged 55-64 have the highest fraud rate (0.49%) and 65+ follow at 0.46% — both significantly above the 0.39% average. The 35-44 cohort is the safest at 0.32%.

**Recommended Action:** Launch a targeted awareness programme for customers 55+ with optional account alerts for any transaction above their personal 90th-percentile spend. Offer a simplified fraud-reporting flow (phone/app) and assign these accounts a lower auto-approval threshold for first-time merchants.

**0.49%**

Fraud rate age 55-64 vs 0.39% avg

**25%+**

Higher fraud exposure for 55+ vs baseline

**~880**

Customers in highest-risk cohort

**P2 MEDIUM**

### Geo-Anomaly Alerts: Flag Transactions Far from Cardholder Home

**Effort: Medium**

**Data Finding:** Fraudulent transactions occur at significantly higher geo-distances from the cardholder's registered address. Merchant locations distant from home are a strong signal when combined with other risk factors such as late-night timing or high amount.

**Recommended Action:** Add a geo-distance feature to the fraud scoring model. Automatically flag transactions where merchant distance > 100km AND the card has had no prior transactions in that state. Send a real-time push notification to the cardholder for approval before settlement.

**100 km+**

Distance threshold for anomaly flag

**2-3x**

Fraud lift expected from geo + time combo

**<30 sec**

Notification latency target for approval

**P3 LOW**

### Increase Fraud Operations Capacity in Alaska, Connecticut & Idaho

**Effort: High**

**Data Finding:** Alaska (1.66%), Connecticut (1.22%), and Idaho (0.88%) have fraud rates 2–4x the national average. Rural areas also show elevated rates (0.42% vs 0.38% urban), suggesting gaps in local fraud monitoring infrastructure.

**Recommended Action:** Assign dedicated fraud analyst coverage to AK, CT, and ID customer portfolios. Lower auto-approval limits in these states by 20% and increase rule sensitivity for transactions originating in low-population zip codes (city\_pop < 5,000). Review merchant-level fraud clustering in Birmingham and Midland for targeted interventions.

**1.66%**

Alaska fraud rate (4x national avg)

**3 states**

Flagged for priority operations review

**~15%**

Estimated fraud reduction in region

## Consolidated Expected Impact

| Recommendation                      | Fraud Loss Prevented | Fraud Rate Impact              | Effort | Timeline  |
|-------------------------------------|----------------------|--------------------------------|--------|-----------|
| Online channel controls             | \$452K/yr            | -40% online fraud              | Low    | 3-4 wks   |
| High-value transaction scoring      | \$689K/yr            | -35% high-value fraud          | Medium | 6-8 wks   |
| 55+ customer programme              | \$95K/yr             | -20% senior fraud              | Low    | 4-6 wks   |
| Geo-anomaly alerting                | \$180K/yr            | -25% cross-geo fraud           | Medium | 8-10 wks  |
| Regional ops uplift (AK/CT/ID)      | \$65K/yr             | -15% regional fraud            | High   | 10-12 wks |
| <b>TOTAL (non-overlapping est.)</b> | <b>~\$1.05M/yr</b>   | <b>-30% overall fraud rate</b> |        |           |

*\* Impact estimates assume independent implementation. Overlapping signals (e.g. high-value + online + geo-anomaly) will amplify detection precision but are conservatively excluded from the total to avoid double-counting. All figures based on Jun-Dec 2020 fraud patterns extrapolated to a 12-month period.*